

Applied Machine Learning: From Fundamentals to Sustainable Applications

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Introduction to ML



Can you recognize him?

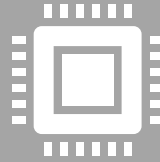


Contents:

- Machine learning vs AI vs Data science
- Supervised vs Unsupervised learning
- Classification and regression
- Steps involved in ML
- Applications
- Introduction to colab by Nussyba



What is ML?



Computer learns from Experience to perform a specific task

Sample Applications:

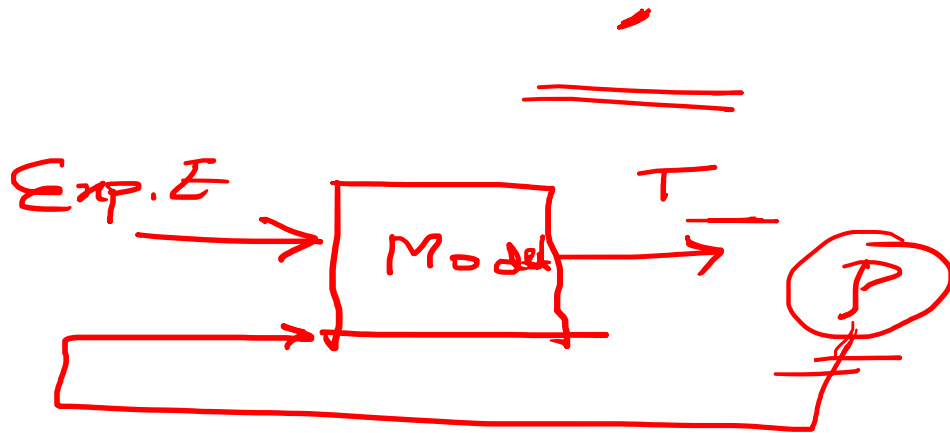
Weather forecasting

Medical applications

Self driving cars

Machine Learning definition

- Tom Mitchell (1998) Well-posed Learning Problem: A computer program is said to *learn* from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

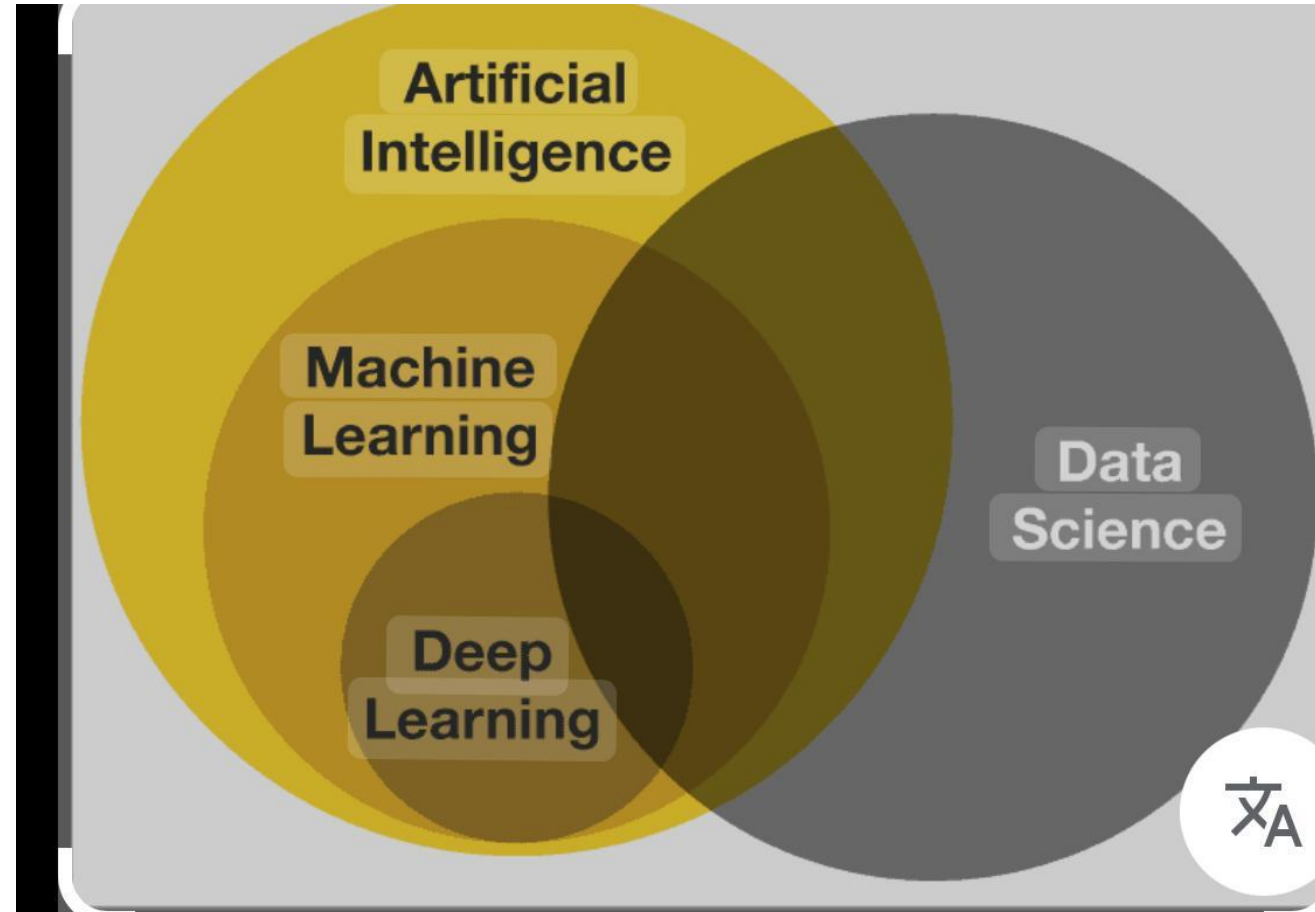


Training (often 70–80% of data) teaches an ML model by letting it learn patterns, while testing (the remaining 20–30%) evaluates the model on new, unseen data to check for accuracy and prevent memorization



Can you recognize him?

AI, ML and DS



- **AI (Artificial Intelligence):** The broadest field, aiming to simulate human intelligence to solve problems. It includes automation, robotics, and reasoning. *Examples: Virtual assistants (Siri), self-driving cars.*
- **ML (Machine Learning):** A subset of AI focused on algorithms that learn patterns from data, rather than being explicitly programmed for every rule. *Examples: spam filters, predictive maintenance.*
- **DS (Data Science):** An interdisciplinary field combining statistics, data analysis, and often ML to find patterns and inform business decisions. *Examples: Data analysis, business intelligence (BI), data visualization, predictive modeling*



Why to study ML

Human expertise does not exist (navigating on Mars)

Models must be customized (personalized medicine)

Working with huge amounts of data



Supervised Machine Learning



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	NT
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	T
	NT
	NT

	X
	X
	X
	X
	X
	X

Vacuum robots (Machine learning?)

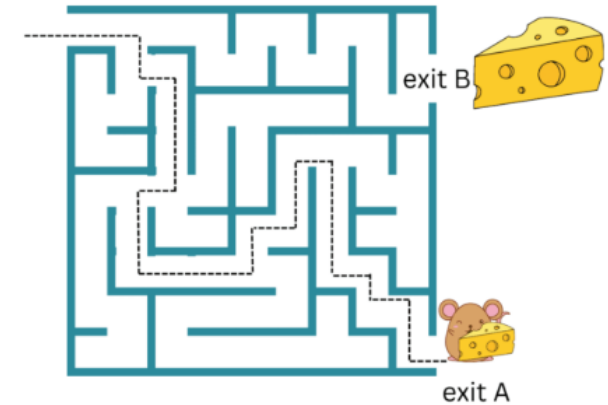


Many vacuum robots just follow rules like:

- Move forward
- If you hit a wall → turn
- If dirt is detected → clean

A robot learns how to clean a maze:

- Trying different paths
- Getting rewards for cleaning
- Getting penalties for collision



Trial and error with rewards

Robots learn **navigation** (finding best path) using Reinforcement Learning

Types of Learning Paradigms

- **Supervised learning**
 - Training data includes desired outputs
- **Unsupervised learning**
 - Training data does not include desired outputs
- **Reinforcement learning**
 - Rewards from sequence of actions
- **Semi supervised learning**
 - Combines supervised and unsupervised learning
- **Weakly supervised learning**
 - learning with noisy labels

Supervised Machine Learning with examples

Toy example: Screening of Mental disorder

Case 1: Answers of two individuals A, B

Q1. Over the last 2 weeks, how often have you been bothered by the following problem?

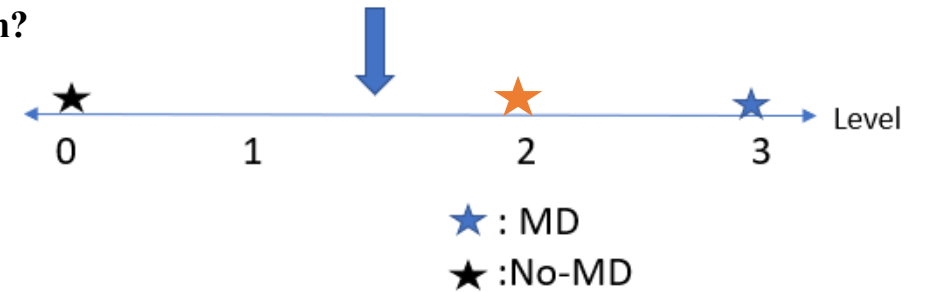
Feeling down, depressed, or hopeless

0. Not at all 1. Several days 2. More than half the days 3. Nearly everyday

ID/Name	Answers to Q1	Category
A	0	No-MD
B	3	MD

Can I use Data of A or B for testing?

$$Th = (3 + 0) / 2$$

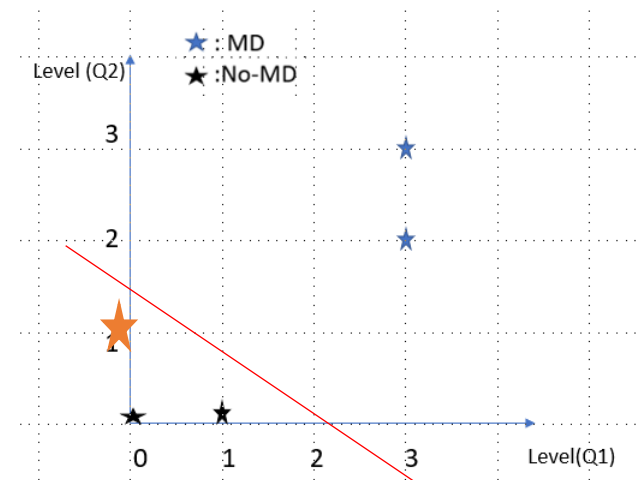


Task 1: If answer of individual C is 2, his category?

Case 2: Answers of four individuals A, B, C, D for two question

Q2. Trouble falling or staying asleep, or sleeping too much

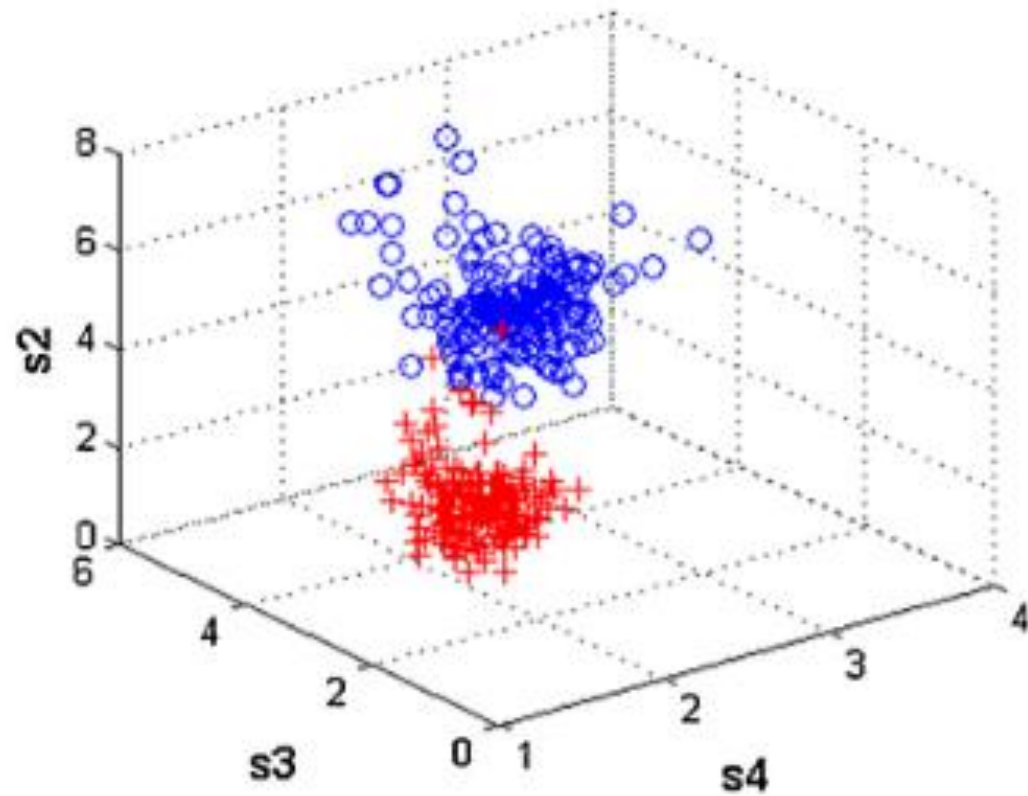
ID/Name	Answers to Q1	Answers to Q2	Category
A	0	0	No-MD
B	1	0	No-MD
C	3	3	MD
D	3	2	MD



Task 2: If replies of individual E is 0 and 1 for Q1 and Q2, his category?

- Data from multiple users: Each answered 3 questions

Nearest neighbor: Searches for closest neighbor
SVM: Line/plane to separate clusters

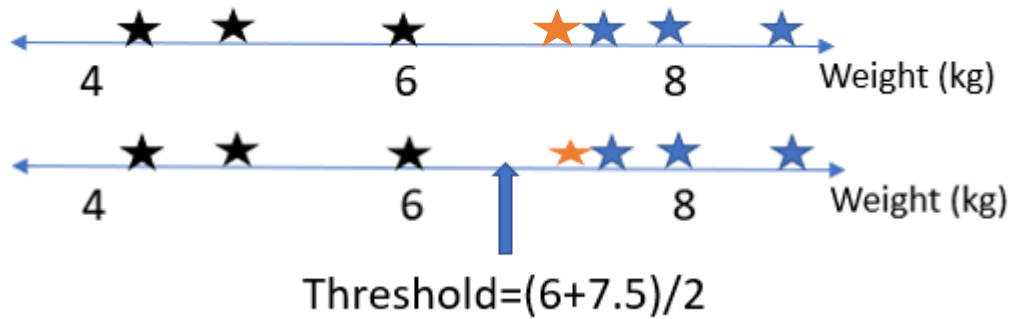


Exercise problem

Example 1: Weight of 3 months boy

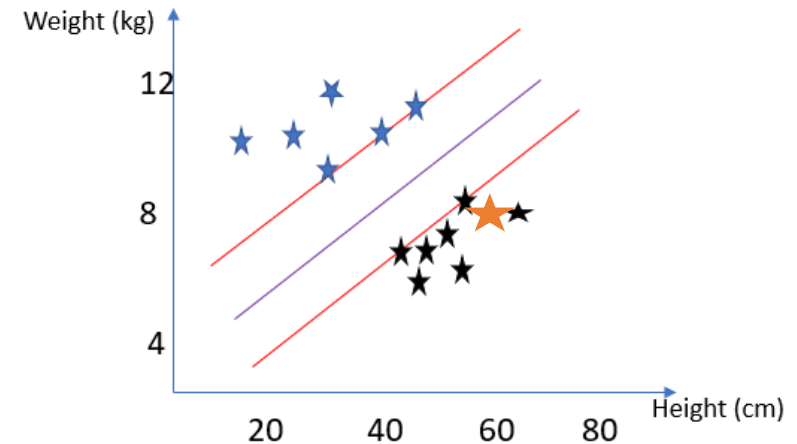
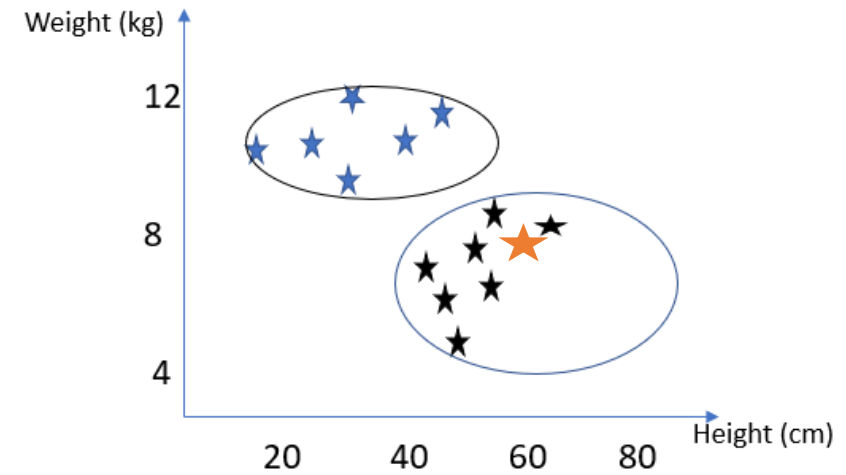
Weight (kgs)	4.4	5	6	7.5	8	9
Label	Others (not-overweight)	others	others	Overweight	Overweight	Overweight

1. If weight=7.3, Category=?



Height	56	58	55	60	70	40	60	6	55	65
Weight	4	4.5	4.6	7	8	8	11	9.5	9	10
Label	Others	Others	Others	Others	Others	Overweight	Overweight	Overweight	Overweight	Overweight

1. If Height=60 cm, weight=6 category=?



Classification predicts categorical classes (like anomaly), while regression predicts continuous numerical values (like age, income, or temperature)

Dependent Variable
(Response Variable)

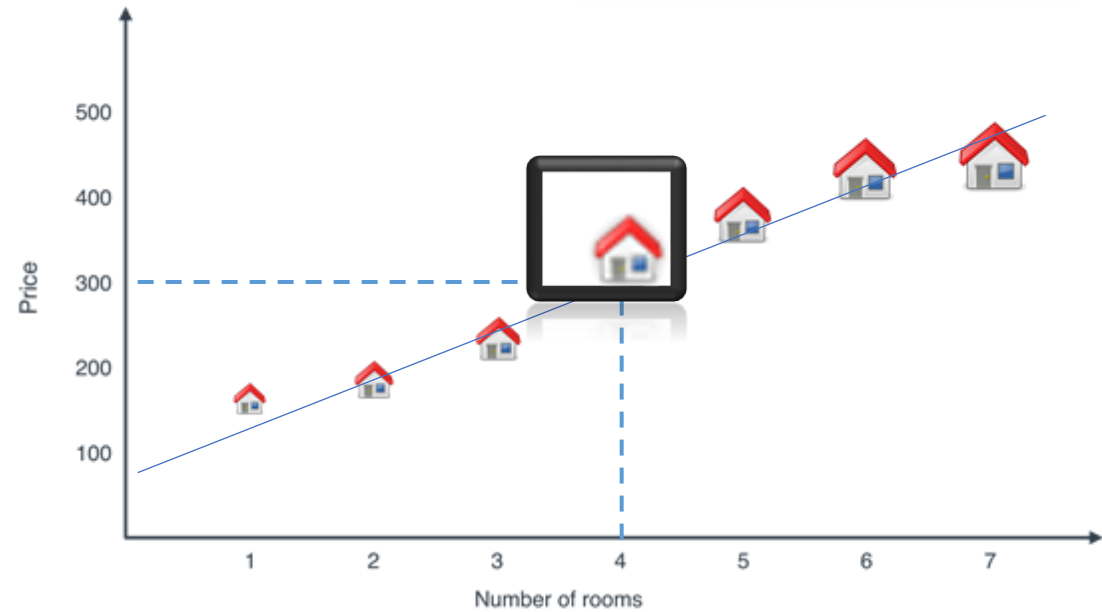
Independent Variables
(Predictors)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots$$

Y intercept

Slope
Coefficient

Rooms=4. Price?



$$y = 100 + 0.5(x)$$

Linear Regression

Example: California Housing dataset

- Predict the house value
- Number of samples (rows): ~20,640.
- Number of features (columns): 8 input features + 1 target variable.
 - **MedInc** – Median income in the block group (in tens of thousands of dollars).
 - **HouseAge** – Median age of houses in the block group (in years).
 - **AveRooms** – Average number of rooms per household.
 - **AveBedrms** – Average number of bedrooms per household.
 - **Population** – Total population of the block group.
 - **AveOccup** – Average number of people per household.
 - **Latitude** – Latitude of the block group (geographical location).
 - **Longitude** – Longitude of the block group

A	B	C	D	E	F	G	H	I
longitude	latitude	housing_m	total_room	total_bedr	population	household	median_income	median_house_value
-122.23	37.88	41	880	129	322	126	8.3252	452600
-122.22	37.86	21	7099	1106	2401	1138	8.3014	358500
-122.24	37.85	52	1467	190	496	177	7.2574	352100
-122.25	37.85	52	1274	235	558	219	5.6431	341300
-122.25	37.85	52	1627	280	565	259	3.8462	342200
-122.25	37.85	52	919	213	413	193	4.0368	269700
-122.25	37.84	52	2535	489	1094	514	3.6591	299200

Typical steps involved

Once we get hold of data, the typical workflow of machine learning are:

- Data acquisition and data cleaning
- Classification/Regression?
- Explore and understand the data (Exploratory Data Analysis)
- Data processing - features extraction, scaling, dimensionality reduction
- Choose Performance metrics
 - i. Classification (Balanced data?)
 - ii. Regression?
- Data partitioning
- model - model selection, hyperparameter tuning
- Assess model – compute model metrics



Data Files

Possible different formats of data files:

- Excel or csv formats (*.xls, '*.csv')
- Fixed width format (*.txt)
- Audio / video / image files
- .mat

Data Cleaning

Eg: Missing data

When we have a dataset that has missing values, we have the options to:

- drop the rows with missing data
- fill in the missing data with estimates

Missing Data

If our dataset is small or too many missing values from different columns, dropping missing data might not be a good option.

- Fill in the missing data with mean of the column
- Fill in the missing data with median of the column

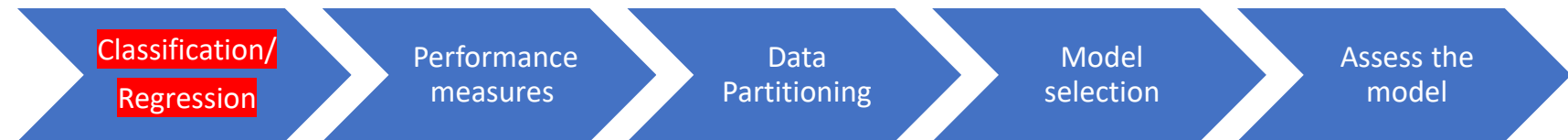
Determine whether the following is classification or regression

Scenario 1.

Feature: Educational background.

Predict if the person is employed or jobless

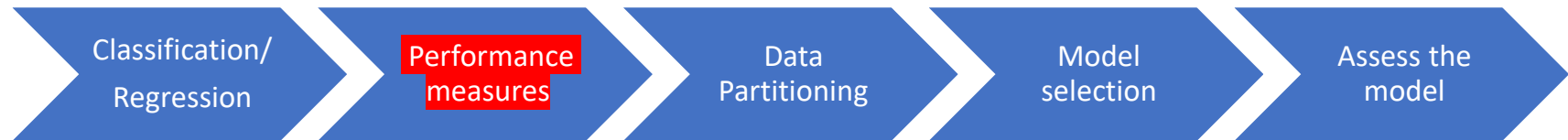
Scenario 2:



Features: Educational background, technical skills

Predict the salary of a given person

Performance Measures



Need for Performance Measure

- To pick one algorithm over another
- To compare different algorithms
- Suitable parameters
- It helps to evaluate the efficiency

- Once you get the output in terms of value or class. The next step is to find out how effective is the model based on some metric using test datasets.

Performance measures (regression)

- Mean Squared Error
- Mean Absolute Error
- Root Mean Square Error

Mean Squared Error

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

MSE = mean squared error
 n = number of data points
 Y_i = observed values
 $\hat{Y}_i$ = predicted values

Mean Absolute Error

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Root Mean Square Error

- $RMSE = \sqrt{MSE}$

Root Mean Square Error

- $RMSE = \sqrt{MSE}$

Performance measures (classification)

- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1-score

Accuracy: Ratio of correctly identified to the total number of test samples

Example

Scenario

We have 100 patients:

- 5 patients have a tumor (positive cases).
- 95 patients do not have a tumor (negative cases).

Assume a model that detected all the cases a no-tumor

Accuracy=0.95%, Is it a good model?

Confusion Matrix

- Confusion matrix is one of the most intuitive and easiest metrics used for finding the correctness and accuracy of the model. It is used for classification problem where the output can be of two or more types of class.

Confusion Matrix		
	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

	Predicted	
	Positive	Negative
Actual	TP	FN
	FP	TN

Truth										
Prediction	Dog	Dog	Dog	No Dog	Dog	No Dog	Dog	No Dog	Dog	Dog
	✓	✗	✓		✓		✗		✓	✗

True **Positive** = 4

False **Positive** = 3

Terminology

- **True Positive:**
 - Interpretation: You predicted positive and it's true.
- **True Negative:**
 - Interpretation: You predicted negative and it's true.
- **False Positive: (Type 1 Error)**
 - Interpretation: You predicted positive and it's false.
- **False Negative: (Type 2 Error)**
 - Interpretation: You predicted negative and it's false.

Accuracy

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

- Ratio of correctly identified to the total number of test samples
- Accuracy can be used for balanced data (equal number of samples)

Precision

$$\textit{Precision} = \frac{TP}{TP + FP}$$

Precision is the percentage of predicted positives that were actually correct

Note: It deals with only predictions

Recall

$$\textit{Recall} = \frac{TP}{TP + FN}$$

=True positives/Actual number of TP

F1-score

$$F1\ score = \frac{2 * Precision * Recall}{Precision + Recall}$$

- Harmonic mean of precision and recall. Penalizes imbalanced performance between precision and recall.
- For imbalance datasets (eg: anomalies, healthcare)
- Arithmetic mean: can be biased towards a high value in one metric

Example

Scenario

We have 100 patients:

- 5 patients have a tumor (positive cases).
- 95 patients do not have a tumor (negative cases).

Assume a model that detected all the cases a no-tumor

Confusion Matrix

	Predicted: Tumor	Predicted: No Tumor
Actual: Tumor	0 (TP)	5 (FN)
Actual: No Tumor	0 (FP)	95 (TN)

Metrics

- Accuracy = $(TP + TN) / \text{Total} = (0 + 95) / 100 = 95\%$ 
- Precision = $TP / (TP + FP) = 0 / 0 \rightarrow \text{undefined (or 0)}$ 
- Recall (Sensitivity) = $TP / (TP + FN) = 0 / 5 = 0\%$ 
- F1-score = 0



Can we have same sample (image here) in training and testing?

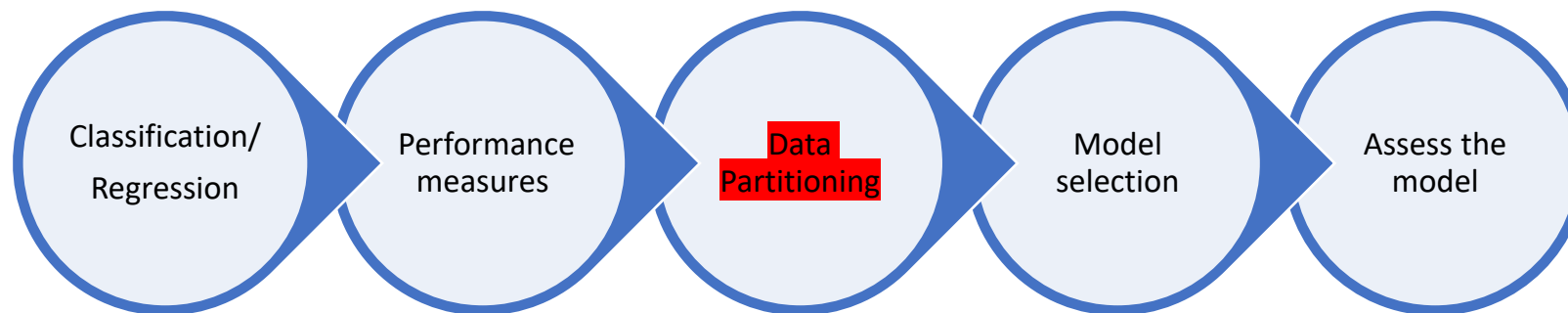


Testing should be performed in unseen data

Depends on the application

- Generally, in healthcare data from same subject cannot be used in test and training
- In face authentication data from same can be used in train and test. But test data should be unseen.

Data Partitioning



Data Partitioning

- To assess how well a model will generalize to unseen data, which is critical for predicting future outcomes.
- Train-Test Validation: Suitable for large datasets where computational resources are not a concern.
- K-Fold Cross-Validation: Ideal for smaller datasets or when you want to get a more accurate estimate of model performance.
- LOSO

K-Fold Cross Validation

- Data is divided into K equally sized portion.
- Each part is used once as the test set while remaining folds form the training set.
- Performance is averaged across all K trials.
- Helps reduce bias due to random sampling.

Eg: K=3, 3-fold

Trial 1

Test 1 (portion 1)

Training (portion 2)

Training (portion 3)

Trial 2

Training (portion 1)

Test 2 (portion 2)

Training (portion 3)

Trial 3

Training (portion 1)

Training (portion 2)

Test 3 (portion 3)

Hold-Out Validation



Train-Test Validation: Suitable for large datasets where computational resources are not a concern.



Data is split into independent training and testing sets (e.g., 70%-30%).



Model is trained on training set and evaluated on test set.



Simple and fast but may suffer from high variance.

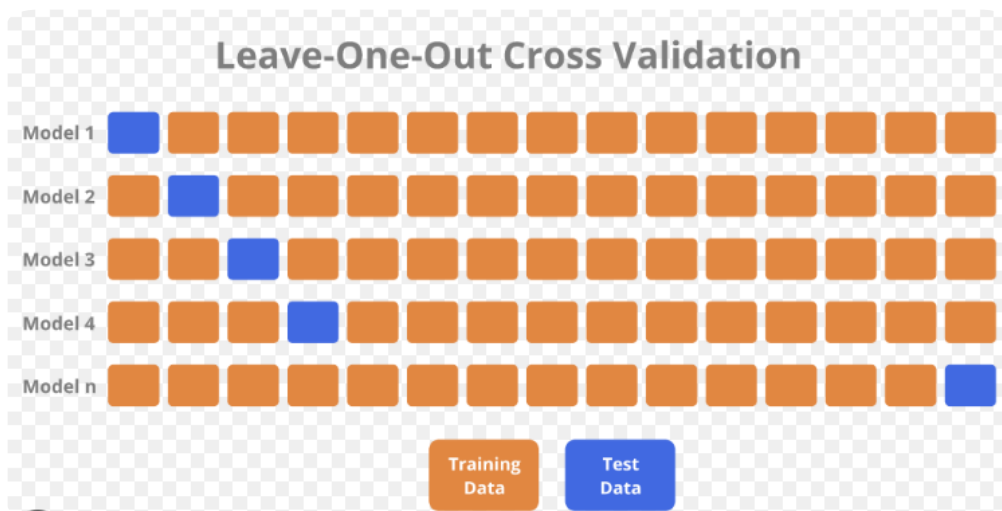


Less robust compared to cross-validation.

Training Set
(70%)

Test Set
(30%)

LOSO (Leave one subject out)

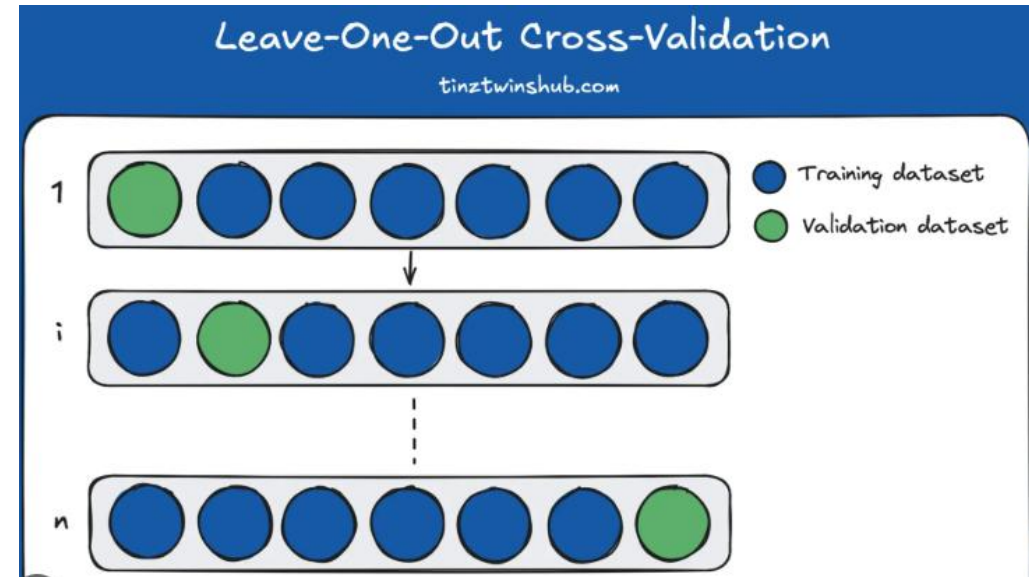


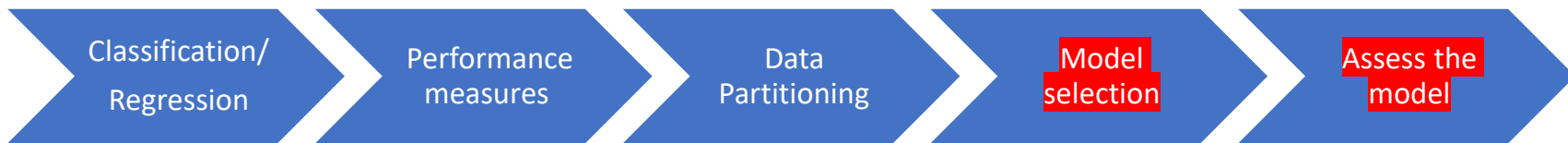
- **Methodology:** If you have data from subjects, the model is trained
• times, each time using subjects for training and the remaining subject for testing.
- **Purpose:** It ensures that data from the same subject does not appear in both the training and testing sets, avoiding "data leakage".
- **Application:** Ideal for small, heterogeneous datasets where traditional
• -fold cross-validation might produce overly optimistic, misleading results.
- **Process:** The final performance is calculated by aggregating the results of all
• iterations
- Mainly used in healthcare.

Leave-One-Out Cross-Validation (LOOCV)

LOOCV creates as many folds as there are data points. Each observation gets its turn as a single-point test set while all remaining observations form the training set

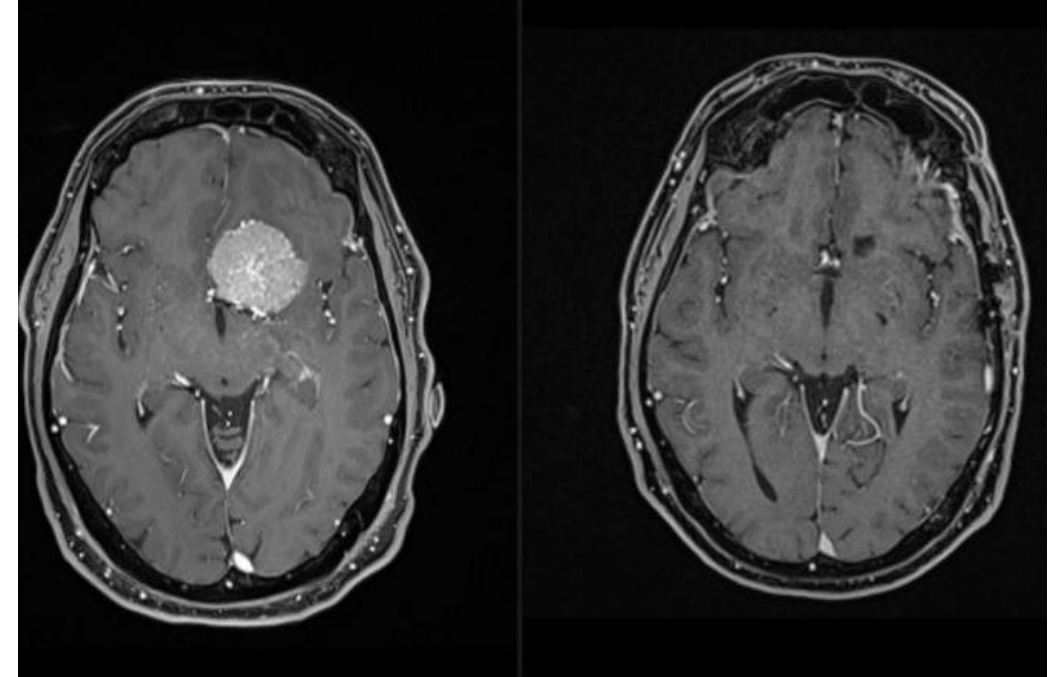
Note: LOSO is at subject level, LOOCV is at sample level. Each subject may contain several samples.





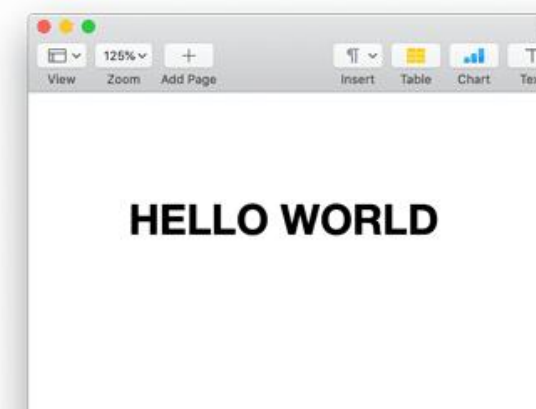
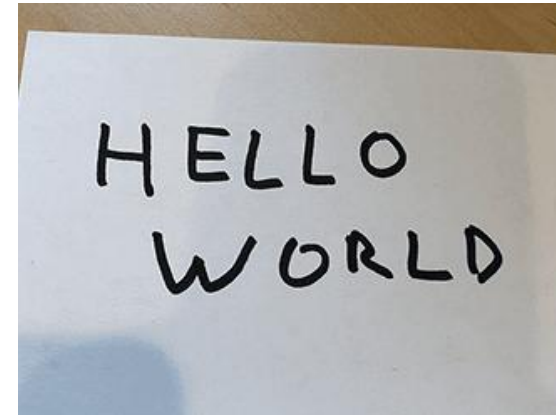
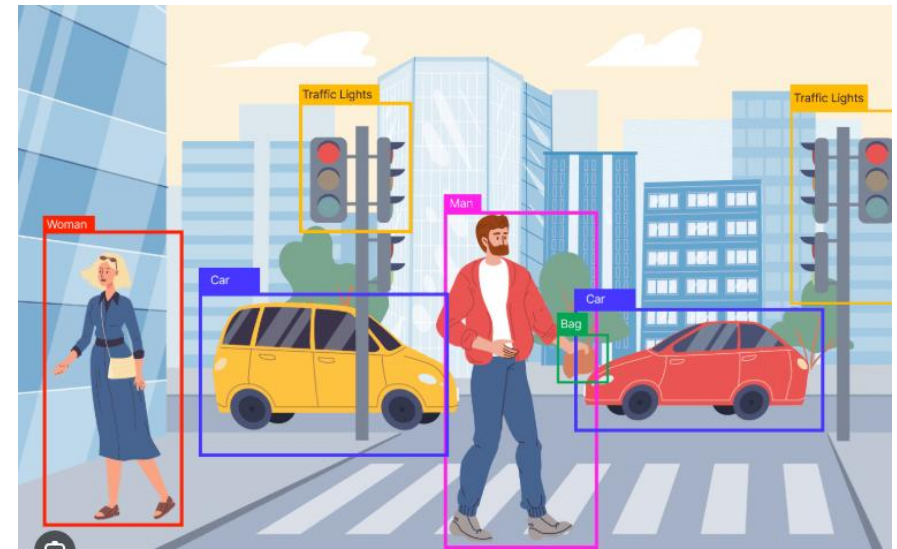
Medicine

- To diagnose a disease
 - Inputs: Symptoms, lab measurements, test results, DNA tests, images
 - Output: healthy or unhealthy, Identify disease
- Employ past medical records to learn and generate a model on which future patients record can be tested



Computer Vision

- ☐ Object Identification
- ☐ Hand-written character recognition (Text to speech for blind)
- ☐ Suspicious activity recognition in surveillance application
- ☐ Face recognition
- ☐ Gender recognition
- ☐ Age Identification



Financial

- ☐ Predict if a stock will rise or fall
- ☐ Predict which ad to show based on individual's past searches
- ☐ Fraud detection: Credit card providers

Online resources:

- Supervised vs Unsupervised Machine learning

[Supervised vs. Unsupervised Learning: What's the Difference in Machine Learning?](#)

- Classification vs Regression

[MFML 030 - Classification vs regression](#)

- Performance metrics:

[How to evaluate ML models | Evaluation metrics for machine learning](#)

[Precision, Recall, F1 score, True Positive | Deep Learning Tutorial 19 \(Tensorflow2.0, Keras & Python\)](#) (with examples)

Thank you